

Measuring Source-Resolution Distortion in LLM Output Arbitration: A Source-Bounded Evaluation Framework

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Dataset: 🤖 <https://huggingface.co/datasets/Ning311/A2R-Benchmark>

Abstract

LLMs are increasingly used to arbitrate among multiple candidate outputs and compress them into a single final result. The final answer may not be incorrect, yet information compression can still make it suboptimal. We define this problem as source-resolution distortion (SRD), the irreversible loss or incorrect mapping of task-relevant source-side structure in the final output during arbitration.

We model pairwise arbitration as a compression process; we then present a source-bounded SRD measurement framework that aligns the reference structure fixed before arbitration with the final output and uses six metrics to measure its preservation and incorrect mapping. To reduce SRD, we further propose A²R (Auditable Arbitration for Resolution), a reference resolution method based on a decision ledger. Motivated by anti-aliasing in signal processing, A²R extracts query-relevant source units and records the local resolution relations among them in a decision ledger before using it to constrain final generation.

Across 520 pairwise cases, compared with Checklist Prompt, A²R-Decision lowers CER from 0.23 to 0.16 and raises Prov-H from 0.45 to 0.67. Its answer-preference scores against three structured baselines range from 51% to 53%, showing that similarly preferred answers can still differ markedly in source recoverability. A 200-case three-candidate extension preserves the system ordering.

1 Introduction

Consider a multi-agent system investigating the following question: “Can a U.S. AI medical-record product enter the Chinese market directly?” One agent argues that market, procurement, and

technical issues can be investigated in parallel; another points out that the regulatory pathway must first be determined by product classification. The arbitration model ultimately synthesizes the two views as follows: “Conduct all investigations in parallel while continuing to confirm the regulatory classification throughout the process.”

This seemingly reasonable decision actually compresses the dependency identified by the sub-agent into a general process reminder. In LLM-as-a-Judge settings, such loss of the original meaning caused by information compression is extremely common. As experiments by Zhang et al. (2025) and Feng et al. (2026) show, the final answer may not be wrong, yet may still be suboptimal because of information compression.

We define the problem caused by this information compression as *source-resolution distortion* (SRD), namely, the irreversible loss or incorrect mapping of task-relevant source-side structure in the final output during arbitration. Existing LLM-as-a-Judge research mostly focuses on (1) judge bias and robustness (Zheng et al., 2023; Shi et al., 2025; Hwang et al., 2025; Halder and Hockenmaier, 2025; Xu et al., 2025a); (2) agreement between LLM judges and human judgments (Liu et al., 2023; Xu et al., 2025b); and (3) evaluative representations and aggregation mechanisms (Lee et al., 2025; Wang et al., 2025; Feng et al., 2026), yet no study explicitly models the resolution distortion caused by the judge.

For this reason, we pose the central question of this paper: when an LLM arbitrates among multiple candidate outputs and compresses them into a single final result, how can we measure the distortion of task-relevant source structure within that result, and how can we reduce this distortion?

Consider again the opening multi-agent example. The problem of SRD is that an origi-

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nally distinguishable signal (product classification determines the regulatory pathway) becomes aliased after aggregation (continue confirming the regulatory classification). For this reason, we draw on the idea from digital signal processing that anti-aliasing filters protect distinguishable components before irreversible downsampling (Crochiere and Rabiner, 1981; Vaidyanathan, 1990). We first fix the pre-arbitration source-resolution structure and use it as a reference to measure the preservation and incorrect mapping of this structure in the final output; we then use blind-reconstruction and decision-utility experiments to test whether this measurement truly reflects structural recoverability. On this basis, we propose A²R, which explicitly preserves source units and their local resolution relations in a decision ledger and then uses the ledger to constrain final generation.

Our contributions are:

- We propose and validate a source-bounded framework for measuring SRD.
- We reveal a decoupling between answer preference and source-resolution fidelity: results with similar preference can still differ substantially in structural recoverability.
- We propose A²R, a decision-ledger-based reference resolution method, and analyze its key mechanisms through ablations.

2 Related Work

Recent work has extensively examined judge bias and robustness, as well as agreement with human judgments. Shi et al. (2025) systematically analyze position bias in pairwise and listwise evaluation. Hwang et al. (2025) find that task-irrelevant persuasive language can systematically alter judge scores, while Haldar and Hockenmaier (2025) reveal substantial self-inconsistency when the same judge repeats an evaluation. ContextualJudgeBench further shows that evaluation relying on external context and conditional criteria remains difficult for current judges (Xu et al., 2025a), while Xu et al. (2025b) find that judge-human correlation declines when candidate systems have similar capabilities. However, these studies mainly characterize the reliability of final judgments and do not trace changes in candidate-side source structure during arbitration.

Another line of work focuses on how evaluation information is organized to form a final judgment. CheckEval decomposes a holistic score into local

checks (Lee et al., 2025); CoT-based Synthesizer uses reasoning traces to integrate complementary information across candidate answers (Zhang et al., 2025); Wang et al. (2025) show that a full judgment distribution outperforms a single final judgment; and Feng et al. (2026) find that implicitly aggregating multiple sub-features produces aggregation inconsistency. These studies aim to improve final evaluation, but do not explicitly measure the loss of source structure during aggregation.

RAG and citation evaluation typically test whether generated claims are supported by, or correctly attributed to, supplied evidence (Gao et al., 2023; Xu et al., 2025c); factuality evaluation asks whether individual claims are correct (Maynez et al., 2020; Min et al., 2023). SRD instead asks whether query-conditioned relations among candidate outputs survive arbitration. A claim can remain source-supported while a conflict is erased or a caveat is detached from the claim it qualifies.

Source separation concerns the recovery of distinct source components from mixed observations and evaluates recovery quality through source leakage and separation distortion (Vincent et al., 2006). In multirate signal processing, anti-aliasing filters protect components that must remain distinguishable before irreversible downsampling, preventing information folding caused by compression (Crochiere and Rabiner, 1981; Vaidyanathan, 1990). These two lines of work inform our modeling and measurement of SRD, as well as the methodological intuition behind A²R. Specifically, we use the pre-arbitration source structure as a reference for measuring post-aggregation recoverability, and A²R explicitly preserves local source relations before holistic generation to reduce the irreversible mixing of task-relevant structure during compression.

3 Problem Definition

3.1 Arbitration Compression

Let q be a query and $Y = \{Y_1, \dots, Y_k\}$ be candidate outputs from LLMs, agents, or retrieval-generation systems. Arbitration maps multiple candidates to a more compact arbitration result:

$$F(q, Y_1, \dots, Y_k) \rightarrow o.$$

Here, o may be a score, a selection result, a synthesized answer, or a structured output with a trace.

This process compresses information. We focus on whether task-relevant source structure is lost or mapped incorrectly during compression.

3.2 Pairwise Arbitration for Modeling

Pairwise comparison is a common basic unit in preference evaluation, reward modeling, and LLM-as-a-Judge (Chiang et al., 2024; Lambert et al., 2025; Zheng et al., 2023). We use pairwise final-answer resolution as the main setting for modeling SRD. For an input

$$x = (q, A, B),$$

the resolver arbitrates between the two candidate outputs according to query q and produces a final answer y and an optional trace τ .

Two sources are already sufficient to produce conflict, complementarity, provenance ambiguity, and caveat migration, and therefore cover the basic forms of SRD. More importantly, the pairwise setting provides a clear resolution boundary. Local relations occur only among a limited set of source units, and the source combinations behind final claims also remain limited. This makes it possible to fix a source-side reference structure before system execution and attribute structural changes in the output to specific source relations.

When the candidate set is extended to a general set of k sources, this boundary quickly becomes more complex. Local relations may form higher-order combinations, and some claims may be authorized only by several sources acting together. The number of possible source subsets grows exponentially with k , while conditions and provenance may span several local relations. The loss of a structure from the final output may then arise from either a local resolution or the subsequent composition of several local resolutions, making both the source-side gold standard and the attribution of distortion difficult to determine uniquely.

The pairwise setting therefore provides an identifiable and attributable basic measurement unit for SRD. The experiments additionally include a $k = 3$ three-candidate check to test whether the main distortion patterns and system ordering remain stable when the locally defined pairwise relations enter a multi-source composition.

3.3 Resolution-Relevant Structure

Definition 1 (Query-conditioned resolution structure). For a pairwise input $x = (q, A, B)$, let

$$S_q(x) = (U_A, U_B, E_q, P_q, Q_q).$$

Here, U_A and U_B are query-relevant source units; E_q records the query-conditioned relations between them; P_q records the provenance information that should remain recoverable for final claims; and Q_q records the caveats that must remain attached to the claims they qualify. $S_q(x)$ specifies what must remain recoverable after arbitration compression, but does not prescribe a unique final wording. The $k = 3$ extension uses the same definition and adds U_C . Complete inventories of relation, operation, and provenance labels appear in Appendices B and C.

3.4 Source-Resolution Distortion (SRD)

A resolver produces SRD when the final answer y , or the output package (y, τ) , fails to preserve elements of $S_q(x)$ or maps those elements incorrectly to final claims.

Definition 2 (SRD decomposition). Let the system output be $(\hat{y}, \hat{L})$, where $\hat{L}$ is an optional ledger or trace, $\hat{U}$ is the set of source units extracted or exposed by the system, and U_q is the union of all candidate source units. SRD can be diagnosed at three points:

$$\begin{aligned} \mathbf{D}_S(\hat{y}, \hat{L}; S_q) = & (D_E(\hat{U}, U_q), \\ & D_L(\hat{L}; E_q, P_q, Q_q), \\ & D_R(\hat{y}; \hat{L}, S_q)). \end{aligned}$$

CER, FCR, Prov-H, and the other checklist metrics instantiate this decomposition during evaluation.

We study five observable forms of SRD:

- **Conflict-handling error.** Conflict erasure occurs when a genuine conflict between sources is not resolved or preserved. False conflict occurs when the system incorrectly creates a conflict between complementary or equivalent sources.
- **Unsupported addition.** The final answer introduces a substantive claim that is not authorized by the candidate sources, query q , or a resolution decision.
- **Provenance loss.** The source of a final claim cannot be recovered or is attributed to the wrong source.

- **Caveat drift.** A condition or limitation is omitted, weakened, or attached to a different target after arbitration.
- **Relation distortion.** A query-conditioned relation between source units is expressed incorrectly in the final output or resolution trace.

For example, “product classification determines the regulatory pathway” in the introduction is a prerequisite in Q_q . Compressing it into “continue confirming the regulatory classification” changes the scope of that condition and therefore constitutes caveat drift.

4 Measurement Model for SRD

The $S_q(x)$ defined in Section 3.3 is the pre-arbitration reference structure. The measurement target is whether its unit relations, provenance assignments, and caveat attachments remain recoverable from the output. For each case, conflict or no-conflict relations, expected provenance, caveat-target pairs, and unsupported traps are fixed from the gold standard before system execution.

Let $\pi_q(s)$ denote the evidence in the final answer or output package that corresponds to structural element s ; if s cannot be recovered, then $\pi_q(s) = \emptyset$. The map π_q describes local correspondence, and the six metrics provide discrete observations. When the gold standard requires a relation between two elements, their provenance distinction, or a caveat’s scope to remain distinguishable in the output, let $\text{sep}_q(s_i, s_j) = 1$. Structural aliasing takes two basic forms: elements that must remain distinct are compressed into the same output component, or the elements are still mentioned but the relation or attachment edge connecting them disappears:

$$\begin{aligned} \text{sep}_q(s_i, s_j) = 1, \quad \pi_q(s_i) = \pi_q(s_j) \neq \emptyset, \\ \text{or} \quad \pi_q(e_{ij}) = \emptyset, \quad \pi_q(s_i), \pi_q(s_j) \neq \emptyset. \end{aligned}$$

Equivalent or complementary content may be merged; $\text{sep}_q = 1$ marks only the relations, provenance distinctions, and caveat scopes that must be preserved. The opening example retains both “product classification” and “regulatory pathway,” but loses the prerequisite relation between them.

The six metrics observe this distortion at different locations. CER measures conflict erasure;

FCR checks false conflict; UAR captures unsupported addition; Prov-H combines provenance exposure and provenance correctness; CPR checks whether caveats remain attached to their targets; and RF* checks the exposure and label fidelity of query-conditioned relations. CER and Prov-H are the primary metrics. FCR, UAR, and CPR are safety diagnostics, while RF* is used only for mechanism analysis. Metric labels, formulas, applicable denominators, and treatment of unclear are given in Appendices B and D.7. Final-only and final-plus-trace outputs are evaluated separately.

In the diagnostic vocabulary of source separation, misattribution corresponds to cross-source leakage, unsupported addition to an artifact, and the disappearance of a relation or attachment edge to structural aliasing (Vincent et al., 2006). The six discrete metrics quantify these distortions separately.

5 A²R: Resolve, Then Aggregate

A²R (Auditable Arbitration for Resolution) makes the local resolution state explicit before final generation. Given (q, A, B) , the system successively produces source units $\hat{U}$, a decision ledger $\hat{L}$, and a final answer $\hat{y}$:

$$(q, A, B) \xrightarrow{E} \hat{U} \xrightarrow{L} \hat{L} \xrightarrow{G} \hat{y}.$$

Under this ordering, arbitration compression is divided across two objects: $\hat{U}$ preserves local information from the candidates that may still affect the answer, and $\hat{L}$ records how that information is combined under the query. Final generation reads only claims already authorized by the ledger. Figure 1 shows the three stages; Y_1, Y_2 in the figure correspond to candidates A, B , and complement corresponds to the relation label additive.

5.1 Source Units and Decision Ledger

The extractor E reads the query and candidate answers and represents content that may license a final claim as

$$u_i = (\text{id}_i, \text{source}_i, \text{text}_i, \text{type}_i, \text{flags}_i).$$

A unit may be a local claim, evidence statement, recommendation, caveat, or risk note. `source` preserves candidate identity, `text` preserves the original anchor, and `type` and `flags` mark functions

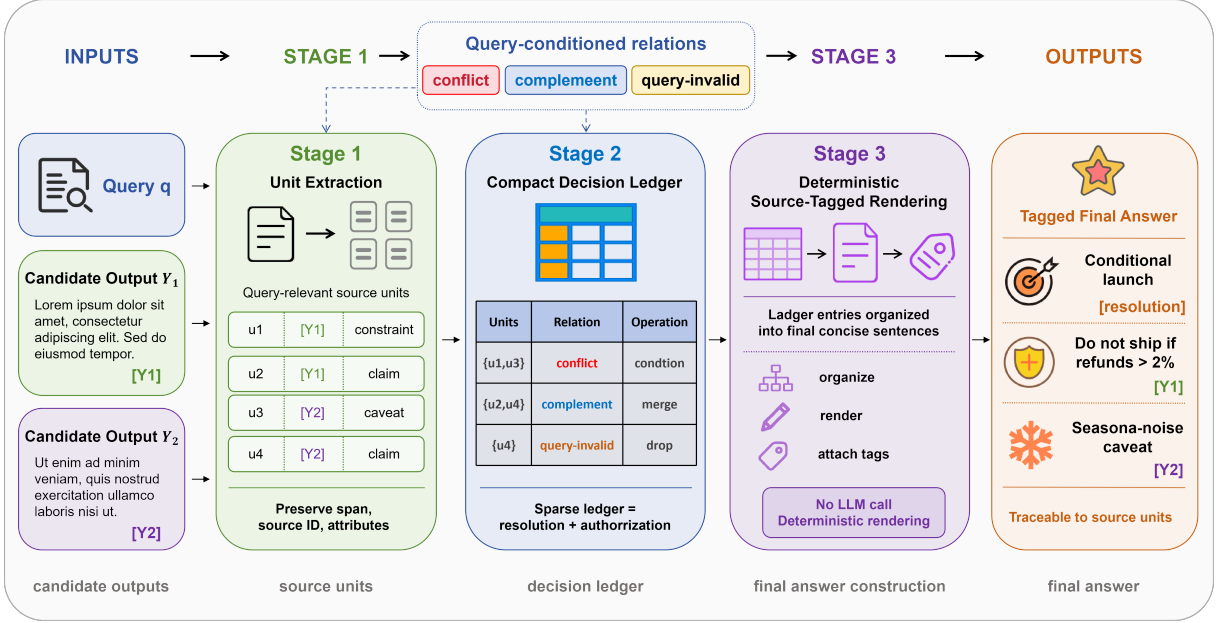


Figure 1: A²R pipeline for source-bounded arbitration.

such as caveat or constraint. E selects query-relevant content; the subsequent decision stage resolves source relations.

The decision module L selects a small set of units from $\hat{U}$ for each local judgment and produces a sparse ledger. Each record is

$$\ell_m = (d_m, I_m, r_m, o_m, c_m, p_m),$$

where d_m is a decision ID, $I_m \subseteq \hat{U}$ contains the units involved in the judgment, r_m is their relation under query q , o_m is the corresponding resolution operation, c_m is the claim licensed for the final answer, and p_m records its provenance category. These components correspond to the decision_id, units, relation, operation, claim, and provenance fields in Appendix C.1.

The relation determines what the ledger can authorize. Equivalent or additive units may be merged, while a query_invalid unit authorizes no final claim. If incompatible units support different conclusions under identifiable conditions, condition_sources retains the conditional alternatives. If the query supplies neither a decision criterion nor sufficient evidence to choose among them, report_unresolved licenses an unresolved statement instead of a synthesized choice. The main comparison uses the merge, condition, and drop operations shown in Figure 1; Section 8.2 evaluates report_unresolved separately on the preference-ambiguous subset. The ledger uses a sparse representation and records the local rela-

tions needed to generate the final answer.

Caveats differ from ordinary claims in how they attach to a conclusion. If a caveat or constraint qualifies a particular conclusion, I_m retains both the caveat unit and its target, and c_m expresses their scope in the same claim. The attachment edge in Q_q thereby enters the ledger.

The authorization relation in the ledger determines the provenance category. Single-source, jointly sourced, query-derived, locally resolved, and explicitly new claims use the [A], [A+B], [query], [resolution], and [new] rules in Appendix C.2. The [resolution] tag marks a claim produced by local resolution and points from d_m to I_m ; in the final-only setting, the provenance category appears directly in the output.

5.2 Ledger-Constrained Deterministic Rendering

The renderer G serializes the claims c_m in the ledger into the final answer:

$$\hat{y} = G(\hat{L}).$$

G uses a deterministic template to order ledger claims, merge adjacent entries on the same topic, and add necessary connectives. Caveat scope, provenance category, and decision outcome are inherited from the ledger. For report_unresolved, the template states the incompatible alternatives and the missing decision criterion without selecting an outcome. Every substantive claim

in the final answer points through `licensed_by` in `final_claim_ledger` to one or more ledger records; these records delimit the claim set of $\hat{y}$.

In the opening example, the extractor obtains two units: “market, procurement, and technical research can proceed in parallel” and “product classification is a prerequisite for determining the regulatory pathway.” The ledger records them with `relation=additive` and `operation=merge`, while preserving the scope of the second unit in the renderable claim. The generated answer is:

```
[r_1][resolution] For non-regulatory matters,
market, procurement, and technical research can
proceed in parallel following Candidate A; given
the prerequisite specified by Candidate B, inves-
tigation of the regulatory pathway should begin
after product classification is determined.
```

The two clauses share decision record r_1 , while the roles of Candidates A and B remain separately recoverable. The product-classification condition still constrains the regulatory-pathway claim. Post-hoc attribution operates on sentences that have already been generated; A²R preserves this attachment before generation.

In the decomposition from Section 3.4, E , L , and G correspond to D_E , D_L , and D_R . The focus of A²R is the processing order: the ledger first preserves the local components that must remain distinguishable, and the generator then performs the tighter final aggregation. This order corresponds to the anti-aliasing principle of protecting distinguishable components before downsampling in multirate signal processing (Crochiere and Rabiner, 1981; Vaidyanathan, 1990).

6 Measurement Validation Design

The six metrics turn SRD into observable checklist labels. Per-metric agreement and the proportion of unclear labels characterize label reliability; blind reconstruction and decision utility further test structural recoverability. Both validation tasks use 50 held-out cases. Table 1 reports their results alongside each system’s Prov-H on the 520-case main set.

Blind reconstruction takes only the system output as input. The task requires recovery of claim provenance, conflict status, caveats and their targets, and unsupported claims from that output. The recovered structure is compared with the source-side structure fixed before system execution to obtain reconstruction accuracy. This task

directly tests the provenance recoverability represented by Prov-H while also observing whether conflict and caveat structures survive in the output.

Decision utility tests whether these structures affect practical use of the answer. For each case, the reference decision, supporting reasons, and blocking risks are fixed before any system answer is viewed. The evaluation input consists of the query and one format-normalized system answer. Normalization removes system-specific bracketed tags, decision IDs, and traces while retaining answer content. The task records decision accuracy, reason coverage, risk coverage, and completion time. Reason and risk coverage are computed against the pre-specified items, while source-resolution labels are used for the structural metrics.

Blind reconstruction measures structural recoverability; decision utility measures decision support and risk identification. We report the two sets of results separately and analyze the associations of Prov-H with reconstruction accuracy and decision accuracy. Assignment and scoring details appear in Appendix D.3, and the numerical results appear in Table 1.

7 Experimental Setup

7.1 Data and Tasks

The main evaluation contains 520 pairwise cases: 320 controlled stratified cases and 200 natural open-ended tasks. The controlled set covers conflicts, provenance, caveats, and unsupported traps, with control strata for no-conflict complementarity, preference ambiguity, weak shared assumptions, and minority-but-critical caveats. The natural tasks come from real user queries, and two external generators independently produce the candidate answers. Source units, query-conditioned relations, expected provenance, caveats, and unsupported traps are fixed before system execution; system identity is hidden during scoring. Applicable item counts, stratified counts, and gold construction appear in Appendices A and D.7.

The query-contrast relation set contains 150 groups that reuse identical or similar source units under different queries to test

$$r = C(u, v; q).$$

The same pair of Kubernetes claims may conflict in technology selection for a small MVP but

System	Main-set Prov-H	Recon. acc. $\uparrow$	Dec. acc. $\uparrow$	Reason cov. $\uparrow$	Risk cov. $\uparrow$	Time (s) $\downarrow$
Free Judge	0.08	0.29	0.42	0.31	0.24	218
Source-Citing Prompt	0.36	0.47	0.53	0.46	0.38	192
Checklist Prompt	0.45	0.58	0.61	0.57	0.51	167
Source-Tagged Synth.	0.52	0.66	0.63	0.61	0.54	153
Post-hoc Attribution	0.55	0.69	0.62	0.64	0.56	161
CoT-based Synth.	0.31	0.41	0.65	0.58	0.46	174
LLM-Blender	0.18	0.32	0.44	0.34	0.27	209
A ² R-Decision	0.67	0.79	0.74	0.71	0.66	128

Table 1: Framework validation on 50 held-out cases; Prov-H is reported on the 520-case main set.

be complementary in an enterprise-platform comparison, separating the effect of the query from textual similarity. The three-candidate check contains 200 (q, A, B, C) cases and uses the same source structure and metric definitions to test whether the system ordering or main distortion patterns reverse. An additional 50 pairwise calibration cases and 10 three-candidate calibration cases are used for development and do not enter the main results.

7.2 Systems and Comparison Protocol

The main comparison contains nine systems. Free Judge directly generates an answer; Source-Citing Prompt requires provenance on each claim; and Checklist Prompt completes a resolution checklist before generating the answer in the same call. Source-Tagged Synthesizer preserves provenance during synthesis, whereas Post-hoc Attribution adds provenance in a second call after an ordinary answer has been generated. CoT-based Synthesizer and LLM-Blender (Jiang et al., 2023) use their public implementations. A²R-Decision uses one LLM call for unit extraction and one for ledger construction, followed by deterministic rendering; A²R-Heuristic replaces only unit extraction with the heuristic in Appendix D.6. Checklist-Templated and A²R-LLM-Final control for deterministic rendering and free-form rewriting, respectively, and appear only in Table 4.

In-task prompt baselines and A²R variants use the same resolver backend and decoding settings, with the candidate set frozen in advance. Model versions, call settings, and prompt packs appear in Appendix D.4. Relative cost is computed from total inference tokens per case with Checklist Prompt set to 1.00; call counts and absolute token use for the structured and multi-stage systems appear in Appendix D.5.

The main metrics are computed on final answers alone. For systems that expose a checklist, trace, or ledger, we also report output-package results.

Metric	Statistic	Value	Unclear
CER	α	0.78	6.2%
FCR	α	0.73	7.1%
Prov-H	α	0.74	8.7%
UAR	α	0.70	7.9%
CPR	α	0.69	10.1%
RF* relation	κ	0.63	12.4%

Table 2: Annotation agreement by metric.

Before answer-preference comparison, bracketed tags, decision IDs, and traces are removed, and the results are summarized as win+tie/2. The relation classifier outputs one label for each (u, v, q) ; the q-blind, shuffled-query, and contrast-query variants change only the query field. Label definitions, prompts, and provenance rules appear in Appendices B and C.

8 Results

8.1 Main Results

Table 2 reports calibrated annotation agreement. CER and Prov-H show the highest agreement and support the main metric analysis. FCR, UAR, and CPR remain interpretable, while RF* relation labels have lower agreement and are therefore used only for mechanism analysis.

Table 3 reports final-only fidelity and relative token cost for the main systems. Free Judge, CoT-based Synthesizer, and LLM-Blender generate fluent holistic or synthesized answers, but have weaker provenance recoverability and higher conflict erasure. Source-Tagged Synthesizer and Post-hoc Attribution provide the closest source-label comparison: they are stronger than ordinary synthesis baselines, but remain below A²R-Decision on Prov-H. A²R-Decision improves CER and Prov-H at a measured cost of 1.44x Checklist Prompt; 1.6x is the pre-specified ceiling in Figure 2.

A²R-Decision improves Prov-H by 0.22 over

System	CER ↓	FCR ↓	UAR ↓	Prov-H ↑	CPR ↑	Rel. cost
Free Judge	0.34	0.05	0.22	0.08	0.58	0.31
Source-Citing Prompt	0.30	0.09	0.18	0.36	0.60	0.46
Checklist Prompt	0.23	0.10	0.15	0.45	0.68	1.00
Source-Tagged Synthesizer	0.22	0.12	0.16	0.52	0.66	1.17
Post-hoc Attribution	0.24	0.09	0.17	0.55	0.64	1.54
CoT-based Synthesizer	0.24	0.06	0.16	0.31	0.64	1.38
LLM-Blender	0.27	0.04	0.16	0.18	0.63	1.22
A ² R-Heuristic	0.19	0.08	0.17	0.60	0.66	0.67
A ² R-Decision	0.16	0.07	0.14	0.67	0.72	1.44

Table 3: Main final-only source-resolution fidelity and token cost.

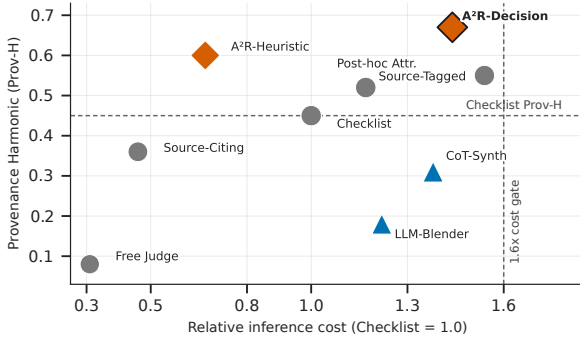


Figure 2: Cost-fidelity frontier for arbitration procedures.

Checklist Prompt and reduces CER by 0.07. Relative to Source-Tagged Synthesizer and Post-hoc Attribution, its Prov-H remains higher by 0.15 and 0.12, respectively. On a 120-case format-normalized subset with bracketed tags and decision IDs removed, A²R-Decision scores 0.60, while Post-hoc Attribution and Source-Tagged Synthesizer score 0.52 and 0.48; Table 4 separately controls for deterministic rendering. A²R-Decision’s FCR is 0.07, below Checklist Prompt and Source-Tagged Synthesizer. UAR is close to Checklist Prompt, and CPR improves only modestly, so we treat them below as safety diagnostics rather than the main gains. A²R-Heuristic is a strong low-cost point: at 0.67x cost, it reaches Prov-H=0.60 and CER=0.19.

Answer preference is evaluated on all 520 pairwise cases after removing system-specific bracketed tags, decision IDs, and traces. Under the win+tie/2 metric, A²R-Decision is close to parity with the structured baselines: 53% against Checklist Prompt, 51% against Source-Tagged Synthesizer, and 51% against Post-hoc Attribution. Rates against all five baselines appear in Appendix D.1.

8.2 Split and Mechanism Checks

Controlled and natural pairwise splits show the same ordering, with smaller gains on natural cases; details appear in Appendix D.2. The main exception is the no-conflict complementary stratum, where the Prov-H gap narrows because few cases require conflict or caveat processing. We also include a 200-case $k = 3$ extension and observe no reversal in system ordering; absolute three-candidate scores appear in Appendix D.13.

Table 4 separates decision structure from the final renderer. Checklist-Templated tests how much deterministic rendering contributes by itself. A²R-LLM-Final tests how much recoverability is lost when the same A²R ledger is rewritten by a free-form LLM. A²R-Decision uses both the decision ledger and the deterministic template.

This ablation separates two contributions. Template rendering reduces provenance loss by restricting free rewriting. The A²R ledger further changes the intermediate object from which the answer is generated, linking final claims to query-relevant units and local resolution decisions.

On the query-contrast relation set, the query-aware classifier reaches a macro-F1 of 0.73, above the q-blind variant at 0.54, the shuffled-query variant at 0.51, and the contrast-query variant at 0.45. The largest gains appear on Conflict and QueryInvalid, showing that arbitration-relevant unit relations are induced by the query and cannot be determined from text similarity alone. Label-level results appear in the appendix. The classifier uses a frozen few-shot LLM prompt and directly outputs one relation label for each (u, v, q) triple; the q-blind and query-perturbed variants change only the query field.

On the 40 preference-ambiguous cases, adding report_unresolved raises correct conditional or unresolved behavior from 52.5% (21/40) to 65.0% (26/40) and reduces incorrectly forced answers

System	Decision structure / Renderer	CER ↓	Prov-H ↑	Rel. cost
Checklist Prompt	Checklist / LLM final	0.23	0.45	1.00
Checklist-Templated	Checklist / Det. template	0.21	0.53	0.95
A ² R-LLM-Final	A ² R ledger / LLM final	0.18	0.58	1.84
A ² R-Decision	A ² R ledger / Det. template	0.16	0.67	1.44

Table 4: Renderer-control ablation.

from 40.0% (16/40) to 27.5% (11/40). The conditional and unresolved breakdown appears in Appendix D.14.

9 Discussion

This paper presents a formal measurement framework and the A²R arbitration method for SRD caused by compression in LLM arbitration. The experimental results reveal a dissociation between answer preference and source-resolution fidelity. A²R-Decision’s answer-preference scores against the three structured baselines are 51–53% under win+tie/2. Free Judge, CoT-based Synthesizer, and LLM-Blender produce fluent answers, but their provenance recoverability (Prov-H of 0.08–0.31) and conflict preservation (CER of 0.24–0.34) are generally weak. One possible reason is that free-form generation tends to smooth over surface disagreements among candidates, losing fine-grained provenance or dependency relations during compression. This gap shows that, in multi-model arbitration and answer-synthesis systems, preference scores and final-text fluency alone are insufficient to reflect the preservation of task-relevant structure. Structural recoverability and textual preference therefore need to be measured separately.

The ablation further shows that reducing SRD requires constraints on both the intermediate decision state and the final generation procedure. In Table 4, using only a deterministic template (Checklist-Templated) or only a decision ledger (A²R-LLM-Final) brings only a partial improvement. A²R-Decision combines the two, raising Prov-H to 0.67 and lowering CER to 0.16. Before generation, the intermediate ledger explicitly records query-conditioned relation judgments and authorizes the corresponding resolution operations, while templated rendering reduces caveat drift and provenance loss caused by free rewriting. The query-contrast experiment also raises macro-F1 from 0.54 for the q-blind classifier to 0.73 for the query-aware classifier, confirming that logical relations among source units are dynamically in-

duced by query context. Separating unit extraction and ledger decisions into stages helps the system identify genuine conflicts and suppress unsupported additions under complex queries.

For deployment, the variants occupy different regions of the cost–fidelity trade-off. A²R-Decision provides the most complete structural preservation and traceable record. Its output package explicitly exposes the ledger, allowing evaluators to inspect the chain of evidence behind each authorization. A²R-Heuristic reaches Prov-H=0.60 and CER=0.19 at only 0.67 times the token cost of Checklist Prompt, providing a practical trade-off for settings with limited computational budgets. These results show that decomposing black-box end-to-end text generation into explicit local resolution and constrained generation can provide a concrete design and evaluation strategy for multi-candidate aggregation systems.

10 Conclusion

This paper formally models and measures SRD and develops an evaluation framework on that basis. The experiments show a clear dissociation between answer preference and source-resolution fidelity: fluent and competitive final answers may still lose critical source structure. Based on this observation, we propose A²R, which extracts query-relevant source units before final aggregation, preserves local resolution relations in a decision ledger, and then performs constrained generation. A²R achieves consistent improvements in CER and Prov-H, while the blind-reconstruction and decision-utility experiments further support the practical significance of source recoverability. The results indicate that the evaluation and design of arbitration systems need to attend to the source structure behind the final answer and preserve critical relations before final generation.

Limitations

Evaluation boundary. SRD is source-bounded: it tests whether arbitration preserves the structure of

the supplied query and candidates, not whether candidate content is externally factual. A source tag or ledger link establishes traceability to candidate content or a resolution decision; it does not establish truth. Retrieval quality, open-world factuality, and real-task success require separate evaluation.

Extraction and unresolved cases. A²R-Decision reaches extraction recall of about 0.91 in the audit diagnostics. A source unit omitted at this stage cannot be recovered by the ledger or renderer. The optional coverage review targets this failure mode at the cost of one additional call, but is not part of the main comparison. The unresolved extension also remains imperfect: it preserves 9 of 18 unresolved cases in the preference-ambiguous subset.

Task scope. The main setting is pairwise final-answer resolution. At $k = 3$, A²R-Decision’s Prov-H changes from 0.67 to 0.55 and its CER from 0.16 to 0.22. Larger heterogeneous source sets, scoring and ranking tasks, and tool-based or iterative multi-agent arbitration are not evaluated here.

Backend and language scope. The main experiments use a closed model backend and a subset check with an open backend from a model family not used for candidate generation. Different backend families may change extraction and ledger error patterns. The data are primarily English tasks; non-English and cross-cultural writing settings require separate validation.

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A Dataset Details

A.1 Calibration Sets

The pairwise calibration set contains 50 controlled cases. It is used only for calibrating the main pairwise task, prompt debugging, and cost sanity checks. Each case includes:

- Query q ;
- Candidate answer A ;
- Candidate answer B ;
- Gold units;
- Gold relations;
- Gold conflicts;
- Gold caveats;
- Expected provenance;
- Unsupported traps.

The three-candidate calibration set contains 10 cases. It is used to check A/B/C source tags, shared-source tags, resolution-derived rules, and deterministic rendering templates, and is kept separate from the three-candidate result set.

Type	Count
Conflict-heavy	40
Provenance-heavy	40
Caveat-heavy	40
Unsupported trap	40
No-conflict complementary	40
Preference-ambiguous	40
Weak shared assumption	40
Minority but critical caveat	40

Table 5: Stratified counts in the controlled diagnostic set.

A.2 Source-Side Gold Construction

Source-side gold structures are fixed before any system output is generated. Gold builders see only the query and candidate answers, not resolver outputs or system identities. They first mark query-relevant atomic source units: claims, conditions, caveats, evidence statements, or recommendations that could license a final-answer claim. Irrelevant background statements are not included unless they change the decision under the query.

The remaining gold fields are derived from these units. A gold conflict is recorded when two applicable units support incompatible actions, conditions, or factual states under the query. A gold caveat is recorded when a condition or limitation must stay attached to a particular conclusion. Expected provenance records which source unit, source set, query condition, or resolution decision should license each final claim. Unsupported traps are substantive claims that would be tempting under surface synthesis but are not licensed by q , A , B , or a grounded resolution decision.

Disagreements are resolved before system execution. Borderline unit boundaries are merged when the same final claim would be licensed either way; otherwise the case is marked for adjudication. Items that remain underspecified after adjudication are excluded from the applicable denominator for that metric rather than assigned to a system after outputs are observed.

A.3 Controlled Diagnostic Set

The controlled diagnostic set contains 320 stratified cases. Each case focuses on a small number of resolution structures and includes negative controls to detect strategies that always add caveats or always report conflicts; the stratified counts are given in Table 5.

A.4 Natural Set

The natural set contains 200 natural pairwise final-answer arbitration tasks. Gold builders see only q, A, B , not system outputs. Output systems do not participate in gold construction.

A.5 Query-Contrast Relation Set

The query-contrast relation set contains 150 groups. Each group reuses the same or similar source units under different queries to directly test query-conditioned relation change.

A.6 $k = 3$ Three-Candidate Check Set

The $k = 3$ check set contains 200 three-candidate final-answer resolution cases. Each case contains a query q and three candidate answers A, B, C , with generation settings matching Section 7. Gold builders see only q, A, B, C and annotate source units, conflicts, caveats, and expected provenance. Output annotators use anonymized system outputs for blind annotation. Each case contains at least two local source relations, allowing us to check the behavior of pairwise-defined local resolution relations when three sources are composed.

The check set uses the same CER, Prov-H, and CPR definitions as the main set. The source set for Prov-H expands from A/B to A/B/C and their combinations. The resolution-derived and query-derived rules remain unchanged.

A.7 Human Annotation Protocol

Human annotators were used for source-side gold construction, output annotation, reconstruction checks, and decision-utility validation. Annotators were internal student annotators and were not paid separately for these annotation tasks. Before annotation, they received the task instructions and label definitions in Appendix B. They were asked to judge only the source-bounded relation between the query, candidate answers, and system output, not the external factuality of the candidate answers. Annotation exports were used only in aggregate for research evaluation. Released artifacts exclude annotator identities, raw annotator logs, account metadata, and free-text notes that could identify an annotator. We did not collect sensitive demographic attributes from annotators.

B Annotation Questions

B.1 CER

Question: Given a gold conflict C_i , does the system output resolve or condition it, or is it erased?

Labels: resolved, conditioned, erased, not_applicable, unclear. Let n_X denote the count of items with label X .

$$\text{CER} = \frac{n_{\text{eras}}}{n_{\text{res}} + n_{\text{cond}} + n_{\text{eras}}}.$$

B.2 FCR

Question: For complementary or equivalent source items labeled as no-conflict in the gold standard, does the system incorrectly report a conflict?

Labels: no_conflict, false_conflict, not_applicable, unclear.

$$\text{FCR} = \frac{n_{\text{fc}}}{n_{\text{nc}} + n_{\text{fc}}}.$$

B.3 UAR

Question: Does the output contain the unsupported trap T_i , or an equivalent substantive unsupported claim?

Labels: absent, present_supported, present_unsupported, unclear.

$$\text{UAR} = \frac{n_{\text{pu}}}{n_{\text{abs}} + n_{\text{ps}} + n_{\text{pu}}}.$$

B.4 Prov-H

Question: Is the provenance of the final claim explicitly recoverable and correct?

Labels: explicit_correct, explicit_wrong, implicit_or_missing, not_applicable, unclear.

Prov-H is formally the harmonic mean of two ratios, similar to F1. The difference is that the two ratios are not ordinary precision and recall, but provenance exposure and provenance correctness for this task. A system may receive an *explicit_correct* label through natural language: for example, “Candidate A gives the latency constraint” counts as explicit correct provenance as long as the source is recoverable and correct.

Coverage and correctness are defined separately. *explicit_wrong* counts in the numerator of P_{cov} because it exposes a checkable provenance attribution; it is then penalized in P_{acc} . This design separates “no provenance exposed” from “provenance exposed but wrong.” The harmonic mean prevents either missing provenance or explicit-but-wrong provenance from receiving a high score

by itself. Let n_{ec}, n_{ew}, n_{app} denote counts of *explicit_correct*, *explicit_wrong*, and applicable items:

$$P_{cov} = \frac{n_{ec} + n_{ew}}{n_{app}},$$

$$P_{acc} = \frac{n_{ec}}{n_{ec} + n_{ew}},$$

$$\text{Prov-H} = \frac{2 P_{cov} P_{acc}}{P_{cov} + P_{acc}}.$$

B.5 CPR

Question: Is the required caveat preserved without substantive distortion?

Labels: preserved, distorted, erased, unclear.

$$\text{CPR} = \frac{n_{pres}}{n_{pres} + n_{dist} + n_{eras}}.$$

B.6 RF*

Question: For this gold-aligned unit pair, what query-conditioned relation does the system expose?

Labels: equivalent, additive, conflict, query_invalid, subsumes, un-related_or_not_comparable, missing_or_not_expressed, unclear.

Let $\mathcal{G}$ be the gold pairs after excluding unclear, and let $\mathcal{E} \subseteq \mathcal{G}$ contain the pairs assigned one of the six relation labels. RF* comprises relation exposure, relation accuracy, and macro-F1:

$$\text{Relation Exposure} = \frac{|\mathcal{E}|}{|\mathcal{G}|},$$

$$\text{Relation Accuracy} = \frac{1}{|\mathcal{E}|} \sum_{i \in \mathcal{E}} \mathbf{1}[\hat{r}_i = r_i].$$

Macro-F1 is computed over the six labels on $\mathcal{E}$. *missing_or_not_expressed* remains in $\mathcal{G}$ but not $\mathcal{E}$; if $\mathcal{E}$ is empty, relation accuracy and macro-F1 are set to zero.

C System Prompts and Schemas

C.1 A²R-Decision Schema

```
{
  "units": [
    {
      "id": "unit_A_1",
      "source": "A",
      "text": "...",
      "type": "...",
      "flags": ["caveat", "constraint"]
    }
  ],
```

```
"decision_ledger": [
  {
    "decision_id": "r1",
    "relation": "conflict",
    "operation": "condition_sources",
    "units": ["unit_A_1", "unit_B_2"],
    "claim": "...",
    "provenance": "resolution_derived"
  }
],
"final_answer": "[r1][resolution] ...",
"final_claim_ledger": [
  {
    "claim_id": "claim_1",
    "text": "...",
    "licensed_by": ["r1"],
    "provenance": "resolution_derived"
  }
]
}
```

The `operation` field also admits `report_unresolved`. Its claim records the incompatible alternatives and the absent decision criterion, and the renderer reports both without choosing an outcome. An optional coverage review may be inserted between extraction and ledger construction. It rereads the query and candidates, appending query-relevant content absent from $\hat{U}$ without making resolution decisions. This review adds one LLM call and is not used in the main comparison.

C.2 Provenance Rules

- Claims supported by a single source use the corresponding source tag, such as [A], [B], or [C].
- Claims supported jointly by multiple sources use shared-source tags, such as [A+B], [A+C], [B+C], or [A+B+C]; [both] in the pairwise task is equivalent to [A+B].
- Claims directly derived from the query use [query].
- Synthesized claims authorized by a grounded decision object use [resolution]. In final-only Prov-H, [resolution] must be accompanied by a recoverable source category. In final-plus-trace Prov-H, it must expose the correct sources or query evidence through the linked decision.
- New inferences that are explicitly marked and reasonable under q use [new].

C.3 Baseline Prompt Skeletons

Free Judge. Given the query and candidate outputs, write the best final answer. Do not output a trace.

Source-Citing Prompt. Given the query and candidate outputs, write the best final answer, and tag each substantive claim with a source tag, [query], or [new].

Checklist Prompt. Given the query and candidate outputs, first answer a compact resolution checklist covering conflicts, caveats, provenance, and unsupported-risk claims, then write the final answer. The checklist and final answer are generated in a single call.

Source-Tagged Synthesizer. First list the query-relevant claims in candidates A and B with source tags, then generate a final answer with [A], [B], [A+B], [query], or [new] tags.

Post-hoc Attribution. The first call generates an ordinary synthesized answer. The second call decomposes the final answer into substantive claims and adds provenance labels for each claim; the final text is not rewritten.

Checklist-Templated. Given the query and candidate outputs, generate only the compact resolution checklist used by Checklist Prompt. A deterministic renderer then converts checklist items into the same source-tagged final-answer template used by A²R-Decision.

A²R-LLM-Final. Run the same A²R unit extractor and decision-ledger generator as A²R-Decision, but replace deterministic rendering with an LLM finalizer asked to write a fluent final answer from the ledger.

C.4 Public-Code Baselines

CoT-based Synthesizer. We use the public implementation of CoT-based Synthesizer as a multi-candidate answer synthesis baseline. We adapt the input format to (q, A, B) for this task without fine-tuning the model; the prompt only adds task wrappers related to the number of candidates and output length.

LLM-Blender. We use the public implementation of LLM-Blender to run candidate ranking and fusion. The candidate set is $\{A, B\}$ for pairwise tasks and $\{A, B, C\}$ for the three-candidate check. PairRanker and GenFuser use the official configuration, with no additional training on our metrics. The public-code version and commit hash are included in the supplementary material.

Baseline	A ² R rate
Checklist Prompt	53%
Source-Tagged Synth.	51%
Post-hoc Attribution	51%
CoT-based Synth.	54%
LLM-Blender	57%

Table 6: Blind answer-preference rates for A²R-Decision (rate = win + tie/2).

Data	N	Δ Prov-H	Δ CER
Controlled pairwise	320	+0.24	-0.08
Natural pairwise	200	+0.19	-0.05

Table 7: Controlled and natural split results.

D Additional Results

D.1 Preference Results

D.2 Split Results

D.3 Framework Validation Details

The 50 validation cases are disjoint from the main set. In reconstruction, readers see only an anonymized output, which is scored against the fixed source structure. In decision utility, readers see the query and one format-normalized answer. System identity is hidden, and a reader never sees two system outputs for the same case. Decision accuracy is agreement with the fixed reference decision; reason and risk coverage are the fractions of pre-specified items recovered. Completion time runs from answer display to submission.

Table 8 reports rank stability across independent annotation pools. The main provenance and conflict metrics preserve system ordering across pools; RF* is less stable and remains a mechanism diagnostic.

Table 9 breaks down the main A²R-Decision vs. Checklist Prompt difference by domain. The effect is largest in product-decision cases and smallest in research-review cases, while the direction is consistent.

D.4 Model Backends

The main resolver backend is OpenAI gpt-5.4-2026-03-05. Candidate answers in the natural pairwise set and the three-candidate check are independently generated before any arbiter is run, using claude-sonnet-4-6, gemini-3.1-pro-preview, and, in the three-candidate check, deepseek-v4-pro. CoT-based Synthesizer and LLM-Blender use the offi-

Metric	A/B	B/C	A/C
Prov-H	0.91	0.88	0.90
CER	0.85	0.82	0.87
FCR	0.80	0.78	0.81
UAR	0.84	0.79	0.82
CPR	0.76	0.72	0.75
RF*	0.61	0.55	0.63

Table 8: Cross-pool rank stability (Kendall’s τ).

Domain	N	Δ Prov-H	Δ CER
System design	142	+0.23	-0.08
RAG/search	89	+0.18	-0.06
Product decision	118	+0.25	-0.09
Research review	95	+0.17	-0.04
Education/planning	76	+0.20	-0.05

Table 9: Domain-level consistency.

cial implementations from their respective papers. The cross-family robustness subset uses Qwen2.5-72B-Instruct, an open resolver backend not used for candidate generation. All main metrics are based on human annotation; model-based evaluators are used only for auxiliary scalability analysis.

D.5 Cost Ablation

The main cost budget is defined over total inference tokens. A²R-Decision uses 1,721 tokens per case, or 1.44x Checklist Prompt, across two serial LLM calls; deterministic rendering adds no model call. The 1.6x line in Figure 2 is the pre-specified cost ceiling. We do not report wall-clock latency. Other cascade, staged, and ledger-first variants—Checklist-Repair, A²R-Checklist-Repair, A²R-Staged, and full A²R-Ledger-First—have relative costs between 1.7x and 4.9x; detailed configurations are provided in the supplementary material.

D.6 A²R Audit Diagnostics

A²R-Heuristic splits each candidate into sentences, ranks them by lexical overlap with the query, and retains at most four units per source. Lexical cues mark conditions, negation, recommendations, and uncertainty; ledger construction and rendering are unchanged. Internal mechanism diagnostics for A²R-style systems show that A²R-Decision has extraction recall of about 0.91, high ledger completeness, and a low invalid-new-inference rate. A²R-Heuristic has substantially lower recall than A²R-Decision, which is the main reason for its lower CPR on minority-critical caveat cases. Full A²R-

Ledger-First is retained only as an upper-bound diagnostic control for the three metrics and is not part of the main comparison.

D.7 Metric Denominators

For all six metrics, unclear is excluded from the denominator; not_applicable, where present, is also excluded. implicit_or_missing remains applicable to Prov-H. If no provenance is exposed, P_{acc} and Prov-H are set to zero. For RF*, missing_or_not_expressed remains in the exposure denominator but is omitted from label-fidelity calculations.

D.8 Final-Only vs. Final-Plus-Trace

D.9 Format-Normalized Provenance Check

On a 120-case subset, we remove system-specific bracketed source tags and decision IDs from the outputs of the three systems in Table 14, keep natural-language source descriptions, and reannotate Prov-H. This check estimates the dependence of Prov-H on explicit tag format.

D.10 Backend Robustness Check

We drew a stratified 120-case subset from the controlled and natural pairwise sets and replaced the resolver backend with Qwen2.5-72B-Instruct. System ordering remains unchanged: A²R-Decision has a Prov-H difference of +0.17 and a CER difference of -0.05 relative to Checklist Prompt. All reported labels still come from human annotation.

D.11 Schema Validation

In the main A²R-Decision run, the JSON schema first-pass success rate is 96.4%. After one retry, the success rate is 99.1%; the remaining 0.9% are marked as invalid ledgers, included in the main final-only outputs, and receive no additional final-plus-trace credit.

D.12 Query-Conditioned Relation Scores

D.13 $k = 3$ Three-Candidate Check Results

This set contains 200 cases and is used to test whether system ordering and main distortion patterns reverse under three-source conditions; see Table 16.

D.14 Unresolved-Case Extension

Before outputs were compared, the 40 existing preference-ambiguous cases were classified from

Component	Backend	temp.	max tok.	calls	retry	prompt pack
A ² R unit extraction	gpt-5.4-2026-03-05	0.0	900	1	1	a2r-v3.1
A ² R decision ledger	gpt-5.4-2026-03-05	0.0	900	1	1	a2r-v3.1
Prompt baselines	gpt-5.4-2026-03-05	0.0	1200	1	0	baseline-v3.1
Post-hoc Attribution	gpt-5.4-2026-03-05	0.0	700	2	0	attrib-v1
Candidate generation	gemini-3.1-pro-preview / deepseek-v4-pro ($k = 3$ only)	0.7	900	1	0	candidate-v2
Open-backend probe	Qwen2.5-72B-Instruct	0.0	900	2	1	a2r-v3.1

Table 10: Main experimental call settings.

System	tokens/case	rel. cost	calls
A ² R-Heuristic	800	0.67	1
Checklist-Templated	1140	0.95	1
Checklist Prompt	1195	1.00	1
Source-Tagged Synth.	1398	1.17	1
LLM-Blender	1458	1.22	2
CoT-based Synth.	1649	1.38	2
A ² R-Decision	1721	1.44	2
Post-hoc Attribution	1841	1.54	2
A ² R-LLM-Final	2198	1.84	3

Table 11: Cost ablation.

Metric	N	Description
CER	512	gold conflict items
FCR	416	no-conflict / equivalent items
UAR	438	pre-specified traps
Prov-H	2184	final claims requiring prov.
CPR	1126	required caveats
RF*	1608	gold-aligned relation pairs

Table 12: Applicable item counts on the main pairwise set.

the query and candidates as conditional (22 cases) or unresolved (18 cases). Each case received two independent labels, with disagreements adjudicated. We compare the original A²R operation set with the same system after enabling `report_unresolved`, while holding the backend, decoding settings, and remaining prompt content fixed. Correct behavior means conditional reporting on conditional cases or preservation of the unresolved state on unresolved cases.

System	final-only	final+trace
Free Judge	0.08	0.08
Source-Citing Prompt	0.36	0.36
Checklist Prompt	0.45	0.58
Source-Tagged Synth.	0.52	0.57
Post-hoc Attribution	0.55	0.61
CoT-based Synth.	0.31	0.39
LLM-Blender	0.18	0.18
A ² R-Decision	0.67	0.83

Table 13: Final-only and final-plus-trace Prov-H.

System	Original (subset)	Tag-stripped
Source-Tagged Synth.	0.51	0.48
Post-hoc Attribution	0.54	0.52
A ² R-Decision	0.66	0.60

Table 14: Tag-stripped Prov-H on a 120-case subset.

Classifier	Macro	Conflict	Additive	QInvalid
$C(u, v; q)$	0.73	0.76	0.71	0.69
$C(u, v)$	0.54	0.50	0.59	0.42
$C(u, v; q_{\text{shuf}})$	0.51	0.48	0.56	0.38
$C(u, v; q_{\text{con}})$	0.45	0.41	0.52	0.35

Table 15: Query-conditioned relation classification.

System	CER ↓	Prov-H ↑	CPR ↑
Checklist Prompt	0.27	0.41	0.65
CoT-based Synth.	0.28	0.29	0.61
LLM-Blender	0.30	0.22	0.60
A ² R-Decision	0.22	0.55	0.66

Table 16: Resolution results on the 200-case $k = 3$ extension.

Outcome	Original A ² R	+ report_unresolved
Correct behavior ↑	52.5% (21/40)	65.0% (26/40)
Incorrectly forced answer ↓	40.0% (16/40)	27.5% (11/40)
Conditional cases handled correctly	—	17/22
Unresolved cases preserved	—	9/18

Table 17: Effect of `report_unresolved` on the 40 preference-ambiguous cases.

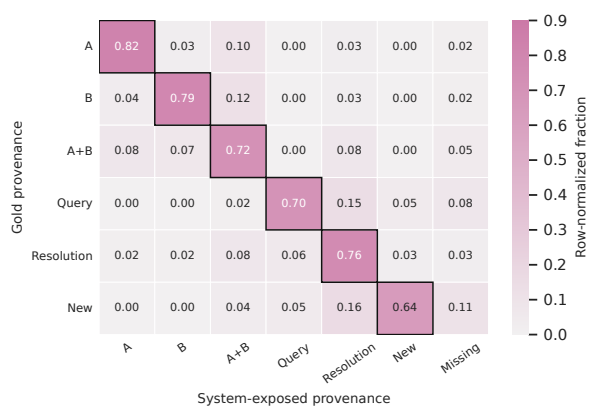


Figure 3: Provenance confusion matrix.